

Motivation

Hallucination

Fluent but factually incorrect responses limit LLM reliability.

Self-Bias

Intrinsic self-correction relies on the same internal knowledge that produced the error.

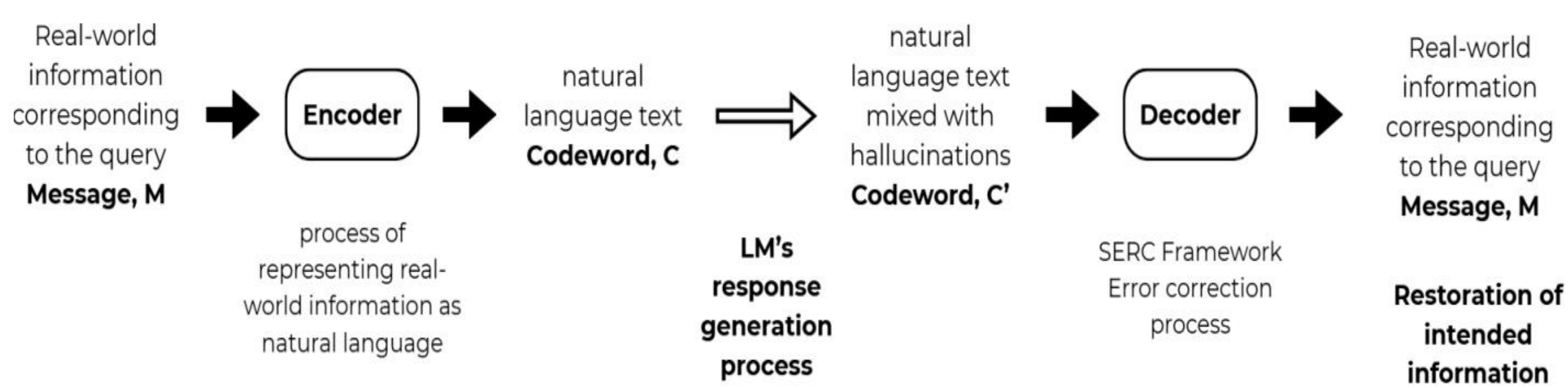
Verification Cost

Claim-by-claim external verification incurs substantial computational overhead.

Purposed Method

Training-free, model-agnostic correction with sparse external verification.

Semantic Noisy Channel

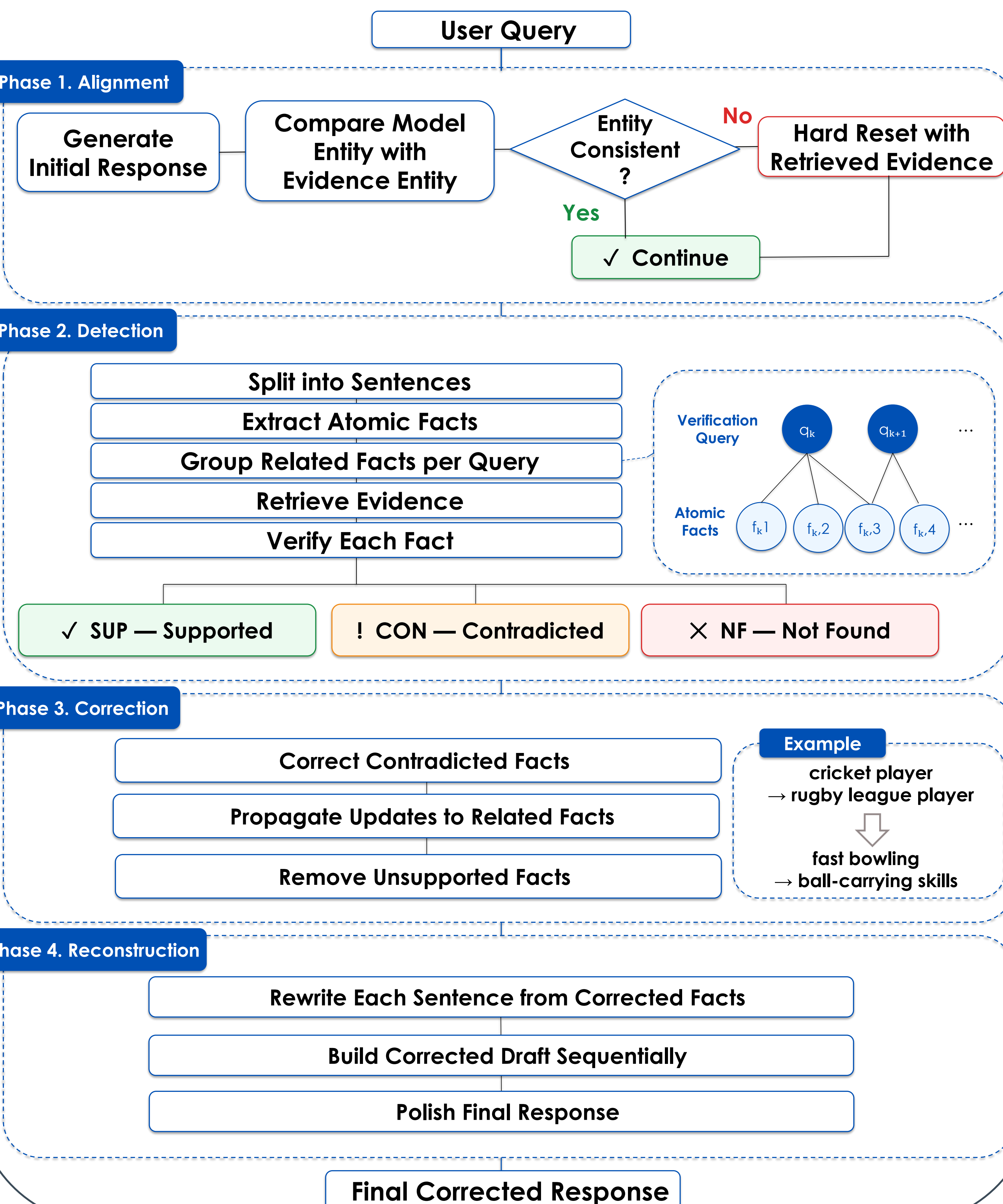


Hallucinations act as semantic noise, producing a corrupted response:

$$C' = C \oplus N_{\text{semantic}}$$

Inspired by error-correcting codes, **SERC** acts as a semantic decoder that detects and corrects corrupted facts using sparse external verification.

SERC FRAMEWORK



Experimental Setup

Backbone Models

- Llama-3-8B-Instruct
- Qwen2.5-14B-Instruct

Benchmarks

- LongForm Bio
- TruthfulQA

Evaluation Metrics

- FactScore
- Fact Preservation Rate
- TruthfulQA Accuracy
- TruthfulQA Score

Full Experimental Details
Prompts, configurations, and additional results are available via the GitHub link in the [More Informations](#) section below.

Main Results

Model	Method	LongForm Bio (Fact Score)	LongForm Bio (Avg Facts)	TruthfulQA (Accuracy)	TruthfulQA (Avg Score)
Llama-3-8B	Initial Answer	0.5976	23.86 (100.0%)	50.0%	5.45
	CoVe	0.5551	17.23 (72.2%)	47.5%	5.45
	CoVe + RAG	0.7265	17.70 (74.2%)	75.0%	7.55
	RARR	0.7999	22.96 (96.2%)	68.5%	7.25
	Re-Ex	0.7636	23.53 (98.6%)	77.5%	7.65
	SERC	0.8568	18.53 (77.7%)	80.0%	8.25
Qwen2.5-14B	Initial Answer	0.4850	21.16 (100.0%)	67.5%	6.67
	CoVe	0.4625	14.85 (70.2%)	60.0%	6.30
	CoVe + RAG	0.7427	17.54 (82.9%)	82.5%	8.47
	RARR	0.7647	21.21 (100.2%)	70.5%	7.65
	Re-Ex	0.7239	21.77 (102.8%)	84.5%	8.53
	SERC	0.8146	22.53 (106.5%)	85.0%	8.62

- SERC achieves the **highest FactScore and TruthfulQA scores** across both backbone models.

Ablation study

Model	Method	Efficiency		Performance	
		Tokens	Reduct.	FactScore	Fact(Pres.%)
8B	Initial	-	-	0.6572	23.55(100.0%)
	High-Density	72,634	-	0.7002	14.50(61.6%)
	SERC	39,492	45.6%	0.8568	18.53(77.7%)
14B	Initial	-	-	0.5050	21.25(100.0%)
	High-Density	54,407	-	0.8128	24.75(116.5%)
	SERC	36,533	32.9%	0.8146	22.53(106.5%)

- Compared with dense one-query-per-fact verification, sparse verification reduces token usage by **up to 45.6%**

Conclusions

Training-Free Framework

SERC restores factual consistency through entity alignment, sparse verification, and BP-inspired logic propagation without additional training.

Improved Factuality

SERC consistently improves factual accuracy across different model scales and benchmarks.

Efficient Verification

Grouped sparse verification reduces the overhead of exhaustive one-query-perfect checking.

More Informations

Github: <https://github.com/labhai/SERC> (QR)

Contact us: {krson, ijang}@hufs.ac.kr



GitHub Page